

Title 1 D;E Identify the study as developing or evaluating the performance of a multivariable prediction model, the target population, and the outcome to be predicted Abstract 2 D;E See TRIPOD+AI for Abstracts checklist Introduction Background 3a D;E Explain the healthcare context (including whether diagnostic or prognostic) and rationale for developing or evaluating the prediction model, including references to existing models 3b D;E Describe the target population and the intended purpose of the prediction model in the context of the care pathway, including its intended users (eg, healthcare professionals, patients, public) 3c D;E Describe any known health inequalities between sociodemographic groups Objectives 4 D;E Specify the study objectives, including whether the study describes the development or validation of a prediction model (or both) Methods Data 5a D;E Describe the sources of data separately for the development and evaluation datasets (eg, randomised trial, cohort, routine care or registry data), the rationale for using these data, and representativeness of the data 5b D;E Specify the dates of the collected participant data, including start and end of participant accrual; and, if applicable, end of follow-up Participants 6a D;E Specify key elements of the study setting (eg, primary care, secondary care, general population) including the number and location of centres 6b D;E Describe the eligibility criteria for study participants 6c D;E Give details of any treatments received, and how they were handled during model development or evaluation, if relevant Data preparation 7 D;E Describe any data pre-processing and quality checking, including whether this was similar across relevant sociodemographic groups Outcome 8a D;E Clearly define the outcome that is being predicted and the time horizon, including how and when assessed, the rationale for choosing this outcome, and whether the method of outcome assessment is consistent across sociodemographic groups 8b D;E If outcome assessment requires subjective interpretation, describe the qualifications and demographic characteristics of the outcome assessors 8c D;E Report any actions to blind assessment of the outcome to be predicted Predictors 9a D Describe the choice of initial predictors (eg, literature, previous models, all available predictors) and any pre-selection of predictors before model building 9b D;E Clearly define all predictors, including how and when they were measured (and any actions to blind assessment of predictors for the outcome and other predictors) 9c D;E If predictor measurement requires subjective interpretation, describe the qualifications and demographic characteristics of the predictor assessors Sample size 10 D;E Explain how the study size was arrived at (separately for development and evaluation), and justify that the study size was sufficient to answer the research question. Include details of any sample size calculation Missing data 11 D;E Describe how missing data were handled. Provide reasons for omitting any data Analytical methods 12a D Describe how the data were used (eg, for development and evaluation of model performance) in the analysis, including whether the data were partitioned, considering any sample size requirements 12b D Depending on the type of model, describe how predictors were handled in the analyses (functional form, rescaling, transformation, or any standardisation) 12c D Specify the type of model, rationale†, all model building steps, including any hyperparameter tuning, and method for internal validation 12d D;E Describe if and how any heterogeneity in estimates of model parameter values and model performance was handled and quantified across clusters (eg, hospitals, countries). See TRIPOD-Cluster for additional considerations‡ 12e D;E Specify all measures and plots used (and their rationale) to evaluate model performance (eg, discrimination, calibration, clinical utility) and, if relevant, to compare multiple models 12f E Describe any model updating (eg, recalibration) arising from the model evaluation, either overall or for particular sociodemographic groups or settings 12g E For model evaluation, describe how the model predictions were calculated (eg, formula, code, object, application programming interface)

Section/topic Item Development/evaluation\* Checklist item Class imbalance 13 D;E If class imbalance methods were used, state why and how this was done, and any subsequent methods to recalibrate the model or the model predictions Fairness 14 D;E Describe any approaches that were used to address model fairness and their rationale Model output 15 D Specify the output of the prediction model (eg, probabilities, classification). Provide details and rationale for any classification and how the thresholds were identified Training versus evaluation 16 D;E Identify any differences between the development and evaluation data in healthcare setting, eligibility criteria, outcome, and predictors Ethical approval 17 D;E Name the institutional research board or ethics committee that approved the study and describe the participant informed consent or the ethics committee waiver of informed consent Open science Funding 18a D;E Give the source of funding and the role of the funders for the present study Conflicts of interest 18b D;E Declare any conflicts of interest and financial disclosures for all authors Protocol 18c D;E Indicate where the study protocol can be accessed or state that a protocol was not prepared Registration 18d D;E Provide registration information for the study, including register name and registration number, or state that the study was not registered Data sharing 18e D;E Provide details of the availability of the study data Code sharing 18f D;E Provide details of the availability of the analytical code§ Patient and public involvement Patient and public involvement 19 D;E Provide details of any patient and public involvement during the design, conduct, reporting, interpretation, or dissemination of the study or state no involvement Result Participants 20a D;E Describe the flow of participants through the study, including the number of participants with and without the outcome and, if applicable, a summary of the follow-up time. A diagram may be helpful 20b D;E Report the characteristics overall and, where applicable, for each data source or setting, including the key dates, key predictors (including demographics), treatments received, sample size, number of outcome events, follow-up time, and amount of missing data. A table may be helpful. Report any differences across key demographic groups 20c E For model evaluation, show a comparison with the development data of the distribution of important predictors (demographics, predictors, and outcome) Model development 21 D;E Specify the number of participants and outcome events in each analysis (eg, for model development, hyperparameter tuning, model evaluation) Model specification 22 D Provide details of the full prediction model (eg, formula, code, object, application programming interface) to allow predictions in new individuals and to enable third party evaluation and implementation, including any restrictions to access or reuse (eg, freely available, proprietary)¶ Model performance 23a D;E Report model performance estimates with confidence intervals, including for any key subgroups (eg, sociodemographic). Consider plots to aid presentation 23b D;E If examined, report results of any heterogeneity in model performance across clusters. See TRIPOD-Cluster for additional details‡ Model updating 24 E Report the results from any model updating, including the updated model and subsequent performance Discussion Interpretation 25 D;E Give an overall interpretation of the main results, including issues of fairness in the context of the objectives and previous studies Limitations 26 D;E Discuss any limitations of the study (such as a non-representative sample, sample size, overfitting, missing data) and their effects on any biases, statistical uncertainty, and generalisability Usability of the model in the context of current care 27a D Describe how poor quality or unavailable input data (eg, predictor values) should be assessed and handled when implementing the prediction model 27b D Specify whether users will be required to interact in the handling of the input data or use of the model, and what level of expertise is required of users 27c D;E Discuss any next steps for future research, with a specific view to applicability and generalisability of the model

Table 3 | Essential items to include for the reporting of prediction model studies in a journal or conference abstract (TRIPOD+AI for Abstracts\*) Section and item Checklist item Title 1 Identify

the study as developing or evaluating the performance of a multivariable prediction model, the target population, and the outcome to be predicted

Background 2 Provide a brief explanation of the healthcare context and rationale for developing or evaluating the performance of all models

Objectives 3 Specify the study objectives, including whether the study describes model development, evaluation, or both

Methods 4 Describe the sources of data 5 Describe the eligibility criteria and setting where the data were collected 6 Specify the outcome to be predicted by the model, including time horizon of predictions in case of prognostic models 7 Specify the type of model, a summary of the model-building steps, and the method for internal validation† 8 Specify the measures used to assess model performance (eg, discrimination, calibration, clinical utility)

Results 9 Report the number of participants and outcome events 10 Summarise the predictors in the final model† 11 Report model performance estimates (with confidence intervals)

Discussion 12 Give an overall interpretation of the main results

Registration 13 Give the registration number and name of the registry or repository